Exercise 1

Task 1: Umbrella world

The set of observable variables $E_t = \{U_t\}$ U - umbrella present at time t.

The set of unobservable variables $X_t = \{R_t\}$ R - rain present at time t.

Dynamic model: $P(X_t \mid X_{t-1})$

$P(R_t \mid R_{t-1})$	$P(R_t \mid \neg R_{t-1})$	0.7	0.3
$P(\neg R_t \mid R_{t-1})$	$P(\neg R_t \mid \neg R_{t-1})$	0.3	0.7

Observation model: $P(E_t \mid X_t)$

$P(U_t \mid R_t)$	$P(U_t \mid \neg R_t)$	0.9	0.2
$P(\neg U_t \mid R_t)$	$P(\neg U_t \mid \neg R_t)$	0.1	0.8

Assumptions in the Umbrella World

1. The 1st order markov assumption
   - Meaning the umbrella world assumes that wether or not it will rain on time t, only depends on wether or not it rained on time t-1.
     For every other time the rain is irrelevant(independant)
2. Conditional independence
   - U_t depends only on R_t and not on U_{t-1} nor R_{t-1}
3. Stationary process
   - Transition and sensor models do not change over time
4. Homogones transistion dynamics
   - Transition probabilities remain constant over time

In [21]:

```
import numpy as np

observation_model = np.array([[0.9, 0.2],[0.1, 0.8]]) # Index 0: U_t tru
sensor_model = np.array([[0.7, 0.3],[0.3, 0.7]]) # Index 0: R_t true, In

belief = np.array([0.5, 0.5]) # No prior knowledge, so assuming equal pr
observations = [0, 0] # No umbrella on both days

def forward(belief, observations):
    new_belief = belief
    for obs in observations:
        prediction = np.dot(sensor_model.T, new_belief)
        obs_prob = observation_model[obs, :]
```

```python
        new_belief = obs_prob * prediction
    # Normalizing beliefs:
    new_belief = new_belief/(np.sum(new_belief))
    return np.round(new_belief, decimals=3)

# --- Task 2a ---

print(f'Normalized probabilities of rain/no-rain on day 2 given observati

# --- Task 2b ---

observations_b = [0, 0, 1, 0, 0]  # Umbrella: true, true, false, true, t

print(f'Normalized probabilities for rain/no-rain on day 5 given the umbr
```

```
Normalized probabilities of rain/no-rain on day 2 given observations of um
brella on both days: [0.883 0.117]
Normalized probabilities for rain/no-rain on day 5 given the umbrella obse
rvations above: [0.867 0.133]
```